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Inventory control under corporate income tax and accrual accounting

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ABSTRACT

Corporate income tax constitutes a significant portion of financial costs in business operations. A typical tax system is characterized by the tax structure, accounting method, and loss carryover rule. In this article, we study the optimal inventory policies under taxation and accrual accounting. We consider a lost-sales periodic-review inventory system under a proportional tax with loss carryforward. We formulate the problem as a cyclic stochastic dynamic program with multiple accounting periods, each of which consists of multiple review periods. We show that an income-dependent basestock policy is optimal. We find that tax function convexity and loss carryforward may introduce conflicting incentives into inventory decisions: the former drives risk-averse decisions whereas the latter induces risk-seeking behavior. We identify two intertemporal effects of taxation, the intra-accounting-period effect and the inter-accounting-period effect, both inducing smaller basestock levels. We further extend the analysis to the progressive tax, backordering system, and cash accounting method. Our numerical study examines the effects of various characteristics of taxation on the optimal policies. Our results demonstrate that ignoring taxation in inventory decisions may lead to significant losses especially when the tax rate and demand uncertainty are sufficiently high, which reveals the value of tax considerations in inventory management.

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Tax policy; inventory control; accrual accounting; stochastic dynamic Program

1. Introduction

Corporate Income Tax (CIT) is a direct tax imposed by a jurisdiction on the operating incomes of a business. It plays a significant role in corporate decisions and profitability. Multinational firms, such as Apple, always carefully structure their investment and sourcing strategies to take advantage of the discrepancies between tax systems (e.g., the *double Irish with a Dutch sandwich*. The recent agreement on the 15% global minimum corporate tax rate between OECD countries indicated an international effort to reduce tax avoidance behavior. Taxation has always been at the forefront of public policy reforms. In the US, the average effective corporate income tax rate in 2017 was 38.9%, the highest one among major economies (NPR, 2017). Prior to 2017, the US federal CIT system consisted of eight tax brackets with a graduated structure, varying from 15% on the first \$50,000 of the taxable income to 35% on the taxable income over \$18,333,333. The 2017 US Tax Cut and Jobs Act (TCJA) reduced the top marginal corporate tax rate from 35% to a flat rate of 21%, aiming to boost US economic growth and international competitiveness. The radical tax reform was believed to have profound economic impacts on US manufacturers to restructure their business operations and supply chains (Ernst & Young, 2018). In response to the tax cut, manufacturers such as Foxconn and a Mazda-Toyota joint venture subsequently announced their plans to build production plants in the US (Isidore, 2018; Shepardson and Woodall, 2018). The IHS Markit US Purchasing Managers' Index indicated a slight increase in the US manufacturing activities in the first three quarters.

A typical CIT system can be characterized by the structure of the tax function, the tax accounting method, the loss carryover rule, and the inventory accounting method.

First, a tax function is characterized by its tax base (i.e., the amount of taxable income to which a tax rate is applied), tax rate, and how the tax rate varies with the tax base (Krugman and Wells, 2017). Although tax structure varies across countries (see, e.g., Deloitte, 2018), it can be classified as being *progressive, proportional, or regressive* in general. In a progressive tax system (e.g., France, Brazil and India), the tax rate increases as the taxable income increases. Progressive tax is most commonly adopted worldwide and is often justified by the ability-to-pay principle. In a proportional or flat-rate tax system (e.g., Hong Kong, Australia and now US), the tax rate is fixed, with no change as the amount of taxable income increases or decreases. Since a firm does not pay tax when the taxable income is negative (i.e., the firm incurs a loss), the proportional tax is a piecewise linear and convex function of the taxable income. The tax function convexity provides incentives for firms to reduce the variation of taxable income in order to reduce expected tax liabilities (see, e.g., Smith and Stulz, 1985; Graham and Smith, 1999).

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Second, there are two basic accounting methods: *cash (basis of) accounting* and *accrual (basis of) accounting*. Under cash accounting, revenues and expenses are recognized when cash is received and paid, respectively. Many small businesses opt to use cash accounting, as it is simple to track cash flows. However, with cash accounting, firms have incentives to purchase or produce excessive stocks to increase their expenses, and thus to reduce tax liabilities. To eliminate such tax avoidance behavior, the IRS discourages the use of cash methods with the exception of some small businesses (Weber, 1961). Under the accrual method of accounting, revenues and expenses are recorded when they occur, regardless of when cash is actually received or paid. The taxable income of a firm is equal to the revenue less the Cost Of Goods Sold (COGS) and other operating expenses. COGS only counts the production or purchase costs of the goods that are issued to meet demand. For businesses with inventory operations, such as manufacturers, firms are not allowed to deduct inventory costs until the inventory is sold. Excess inventories carried over to the future periods are treated as current assets, and the production or purchase costs are deferred until they are sold. Accrual accounting matches expenses with revenue, and hence, costs associated with the revenue of a period need to be properly allocated to that period no matter when they are actually incurred. An item manufactured or purchased for later sale stays on the balance sheet as inventory under current assets until it is actually sold or becomes obsolete; at that point it will go on the income statement under COGS. Under accrual accounting, firms will not intentionally hold excessive inventories. Accrual accounting is considered to be the standard accounting practice for most companies, and is required for firms under both the International Financial Reporting Standards (IFRS) and the U.S. standard of Generally Accepted Accounting Principles (GAAP) (IRS, 2019). In particular, U.S.-based publicly traded companies are required to file financial reports with the Securities and Exchange Commission under GAAP accounting (Gnanarajah, 2014). Such a reality has long been recognized in both accounting and inventory literatures (see, e.g., Cohen and Halperin (1980), Biddle and Martin (1985, 1986) and Yan *et al.* (2019) and references therein). To capture the reality that both IFRS and GAAP mandate the use of accrual accounting, we formulate the inventory control problem under accrual accounting.

Third, many tax systems allow tax payers to carry over operating losses to the future (or past) years to offset tax liability. TCJA removed the previous two-year carryback provision, extended the 20-year carryforward provision out indefinitely, and limited carryforwards to 80% of net income in any future year. Tax loss carryforward allows firms to spread losses over time, inducing *risk-taking* behavior (Feldstein, 1969; Langenmayr and Lester, 2018).

Fourth, there are two alternative methods for inventory accountability: First In First Out (FIFO) and Last In First Out (LIFO) (IRS, 2019). Whereas GAAP permits the use of both methods, IFRS only allows FIFO accounting. Cohen and Halperin (1980) and Biddle and Martin (1985, 1986) have studied optimal inventory policies under FIFO and LIFO.

Due to the significance of tax liabilities, inventory management under CIT in dynamic and stochastic environments requires decision makers to explicitly account for the underlying tax structures and accounting methods. However, the standard stochastic inventory models (see, e.g., Porteus (2002) and references therein), though with a few exceptions, which focus on maximizing expected profits or minimizing expected costs without considering tax liabilities, may lead to suboptimal solutions that will not maximize the expected post-tax profits. To fill this gap between inventory theory and real-world practice, we address the following research questions: What is the optimal inventory policy under CIT? What is the effect of tax structure on the optimal policies? And how do the conflicting effects induced by tax function convexity and loss carryforward interplay with each other?

In our main model, we consider a periodic-review inventory system with stochastic demand and lost sales under taxation. We capture three important characteristics of CIT: the accrual accounting method for firms with inventory operations, tax function convexity under a progressive tax, and the loss carryforward rule. The variable ordering cost is fixed throughout the planning horizon, which implies that there is no difference between FIFO and LIFO accounting. A firm maximizes its expected discounted after-tax profit over multiple accounting periods, each of which consists of multiple replenishment periods. The problem is formulated as a cyclic stochastic dynamic program. We show that the optimal replenishment policy can be characterized by a state-dependent basestock policy. To gain insights on the intertemporal effects of taxation, we consider two special settings: a single accounting period, which consists of multiple ordering periods to account for the intra-accounting-period effect of taxation, and multi-accounting periods where each of which consists of one ordering period to account for the inter-accounting-period effect. In both settings, we show that the basestock levels under taxation are less than those without taxation. In the numerical study, we systematically examine the effects of different characteristics of taxation and the value of considering taxation in the inventory decisions. We also extend the structural analysis to the system under a progressive tax, the backordering case, and the cash accounting method.

Our contribution is twofold. First, to the best of our knowledge, this article is the first one that simultaneously captures the key characteristics of the real-world tax systems: the accrual accounting method and loss carryforward rule, which extends the current literature on inventory control under taxation. We characterize the structure of the optimal policy for both lost-sales and backordering systems. Our numerical results show that the potential loss of the expected post-tax profit from ignoring tax liability in inventory decisions can be significant, especially when the tax rate and coefficient of variation (cv) are sufficiently large. Second, our analysis for the effects of CIT (e.g., the tax function convexity and the loss carryforward) on the optimal inventory policies sheds new light on the conflicting

incentives induced by different accounting rules of the tax systems.

The remainder of this article is organized as follows. Section 2 reviews related literature. Section 3 offers a general dynamic model with consideration of CIT. Section 4 derives some structural properties and characterizes the optimal policy. Section 5 examines the effects of taxation. Section 6 numerically studies the effect of tax and quantifies the profit loss due to ignoring tax. Section 7 concludes with the key managerial insights. Extensions to the system under a progressive tax, the backordering system, and the cash accounting method, and all proofs are in the Supplement Online Materials.

2. Related literature

Our work is related to economics and accounting literature on taxation, see, e.g., Graham and Smith (1999); Sarkar and Goukasian (2006) and Sarkar (2008). For example, Djankov *et al.* (2010) empirically show that the effective corporate tax rates have a large adverse impact on the aggregate investment. Eldor and Zilcha (2002) and Zilcha and Eldor (2004) study the effect of tax asymmetry (i.e., the tax rate on profits is higher than the subsidy rate to losses) on a risk-neutral firm's production and hedging behavior under price uncertainty, and find that tax asymmetry lowers the firm's optimal output. Langenmayr and Lester (2018) show that tax rates and tax loss rules are critical when assessing how tax policies affect firms' investment decisions. Different from their focuses on firms' capital structure and financial decisions, we examine how tax affects firms' inventory decisions.

There are a few studies on dynamic inventory policy under taxation. A stream of work focuses on the effect of tax on inventory policies under different inventory costing methods, such as LIFO, FIFO, and average cost. Cohen and Pekelman (1979) is the first to study the fundamental problem: How should a firm manage its inventory under tax liabilities in systems with stochastic demand and changing costs? They analyze the optimal inventory policies under FIFO and LIFO in a lost-sales system under a flat-rate tax. In the one-period setting under mild assumption of cost inflation, they show that LIFO inventory levels tend to be higher than FIFO inventory levels, and an increase in the tax rate under LIFO [FIFO] may lead to an increase [a decrease] in the inventory. In the multi-period setting under FIFO, they show that a myopic basestock policy is optimal under two critical assumptions: (i) demand is always greater than the starting inventory level of each period; and (ii) the taxable income of each period is equal to the sales revenue less the cost of goods sold. The first assumption implies that all old inventories are consumed, which greatly simplifies the analysis by restricting the inventory ages to two at maximum. The second assumption ignores the operating expenses (e.g., holding cost) in the taxable income, and ensures that the taxable income is always positive and therefore the tax function is linear in the taxable income under a flat-rate tax. In a follow-up work, Cohen and Halperin (1980) develop a deterministic model to examine both intra-year and end-of-year inventory decisions with nonstationary demand under LIFO and illustrate the interplay between the intra-year and end-of-year decisions. Biddle and Martin (1985) and Biddle and Martin (1986) extend Cohen and Halperin (1980) to a stochastic model where a firm can place an initial order at the beginning of each year and a year-end order after demand realization, which allows for an explicit consideration of year-end tax effects. All these papers are based on accrual accounting. Complementary to this stream of literature, we develop a more general model with multiple accounting periods under accrual accounting and investigate the impact of key characteristics of taxation (e.g., tax function convexity and loss carryforward) on the inventory policies.

Although both the international standard of IFRS and the U.S. standard of GAAP mandate the adoption of accrual accounting for firms involving inventory assets, there remains some theoretical interests to study the inventory policies under cash accounting; see, e.g., Xue (2015) for a study on inventory policies under cash accounting in a single-accounting-period setting without considering loss carryover. In addition to the difference in the accounting method, we address another key characteristic of CIT, the loss carryover policy, which requires us to formulate the problem over multiple accounting periods. The structural analysis of such a new model turns out to be more challenging than the single-accounting-period setting. We identify a sufficient condition under which the state-dependent basestock policy is optimal.

More generally, our work falls into the area of the interface of tax and operations. A stream of work studies the operational decisions of a multinational firm that operates across different tax jurisdictions. Shunko and Gavirneni (2007); Huh and Park (2013) and Shunko *et al.* (2014) consider transfer pricing, which is one of the main mechanisms for allocating profit among different divisions of a firm. Xiao *et al.* (2015) study capacity decisions in the subsidiary of a multinational firm which uses foreign tax credits to offset their tax liabilities and avoid double taxation. These papers focus on the impact of tax rate difference on a multinational firm's transfer pricing and operational decisions in a static setting. Different from these works, we focus on the effects of taxation on dynamic inventory polices.

Since tax function convexity, which results in concavity of the post-tax income, plays a similar role of managerial risk aversion, our model is also related to the literature on risk-averse inventory models (see, e.g., Eeckhoudt *et al.*, 1995; Agrawal and Seshadri, 2000). In particular, Eeckhoudt *et al.* (1995) show that the risk-averse newsvendor solution tends to be less than the risk-neutral one. In a dynamic setting, Bouakiz and Sobel (1992) study inventory replenishment with an exponential utility criterion and show that a basestock policy is optimal. Chen *et al.* (2007) propose a framework that incorporates risk aversion in multi-period inventory models and study a firm's inventory and pricing strategies under an additive exponential utility. They assume that the "additive independence axiom" holds, i.e., the utility of the decision maker is the sum of the utility in each period. They show that the (s, S, p) policy is optimal.

Kouvelis *et al.* (2018) study a dynamic financial hedging and inventory control problem in a mean-variance framework and show that a state-dependent basestock policy is optimal with a smaller optimal basestock level in comparison with its risk-neutral counterpart. In our model, the firm's expected after-tax profit in an accounting period can be viewed as a utility function of the sum of taxable income of each ordering period that is not equal to the sum of after-tax profit of each ordering period. In this case, the "additive independence axiom" no longer holds, which imposes technical challenges to the analysis. It is notable that the loss carryforward rule tends to offset the convexity of tax function and induce risk-seeking behavior, which further complicates the analysis. In addition, the dynamic risk-averse inventory models focus on the characterizations of optimal policies and they do not further explore the effects of risk aversion on the optimal policy. Furthermore, most existing inventory models use cash-basis accounting with an exception of Yan *et al.* (2019) who examine a multi-period inventory model with a chance constraint in each period that requires meeting a profit target with a given probability. Our work complements to this stream of literature by addressing the effect of taxation.

From a methodological perspective, our work is also related to the literature on inventory management with cash constraints. Chao *et al.* (2008) consider a self-financing multi-period stationary lost-sales inventory system with zero holding cost and show that a capital-dependent basestock policy is optimal. Gong *et al.* (2014) generalize Chao *et al.* (2008) to a system with limited capital and short-term financing. Assuming an increasing convex loan rate, they show that a capital-dependent basestock policy is optimal. In a similar setting, Katehakis *et al.* (2016) consider a piecewise linear transaction cost function for both financing and cash investment and show that the optimal policy can be characterized by multiple state-dependent thresholds. Recently, Fu *et al.* (2021) study the inventory problem of a capital-constrained firm under an inventory-based financing scheme. Assuming zero holding cost, they show that it is optimal to order up to a state-dependent order-up-to level subject to its capital and loan constraints, or salvage existing inventory down to a state-dependent salvage-down-to level. In inventory systems with taxation, tax can be viewed as an alternative financial cost that is imposed on the operating profit generated in each accounting period, which explains certain similarities in the formulations and structures of the optimal policy between our model and the cash-constrained models. The key difference of our model from the above cash-constrained models is that tax is paid on the taxable income accumulated in each accounting period, which depends on the tax policy such as accounting method (i.e., accrual or cash) and loss carryforward rule. Such difference poses new analytical challenges and leads to different structural properties and new managerial insights as a complement to these literature.

3. The model

Consider a firm managing a periodic-review inventory system with lost-sales over N accounting periods (e.g., fiscal years), indexed by $t = 1, \ldots, N$. Each accounting period consists of M review periods, indexed by $m = 1, \ldots, M$. Period (t, m) refers to the mth review period within the tth accounting period. The discount factor for each accounting period is denoted by $\gamma \in (0, 1]$.

At the beginning of period (t, m), the firm observes the on-hand inventory level $x_{t,m}$ and the accumulated taxable income $w_{t,m}$ of the current accounting period. The firm decides the order-up-to level $y_{t,m}$ $(y_{t,m} \geq x_{t,m})$ at a unit ordering cost c which is constant over time. The random demand, denoted by $D_{t,m}$, is realized during the period. We assume that demands are independently distributed over time and $D_{t,m}$ has a continuous support $[0, +\infty)$ with the cumulative distribution function $F_{t,m}(\cdot)$ and the density function $f_{t,m}(\cdot)$. The unit selling price is p with $p > c$, i.e., the marginal profit is non-negative. We use $x^+ \equiv \max\{x, 0\}$. Excessive inventory, $x_{t,m+1} = (y_{t,m} - D_{t,m})^+$ for $m < M$ or $x_{t+1,1} = (y_{t,M} - D_{t,M})^+$, is carried over to the next period, incurring a unit holding cost h per period, and the unmet demand is lost. The operating profit obtained in period (t, m) is

$$\varphi_{t,m}(x_{t,m}, y_{t,m}, D_{t,m}) = \tilde{\pi}_{t,m}(y_{t,m}, D_{t,m}) + cx_{t,m}, \tag{1}$$

where

$$\tilde{\pi}_{t,m}(y_{t,m}, D_{t,m}) = p\min(D_{t,m}, y_{t,m}) - cy_{t,m} - h(y_{t,m} - D_{t,m})^+. \tag{2}$$

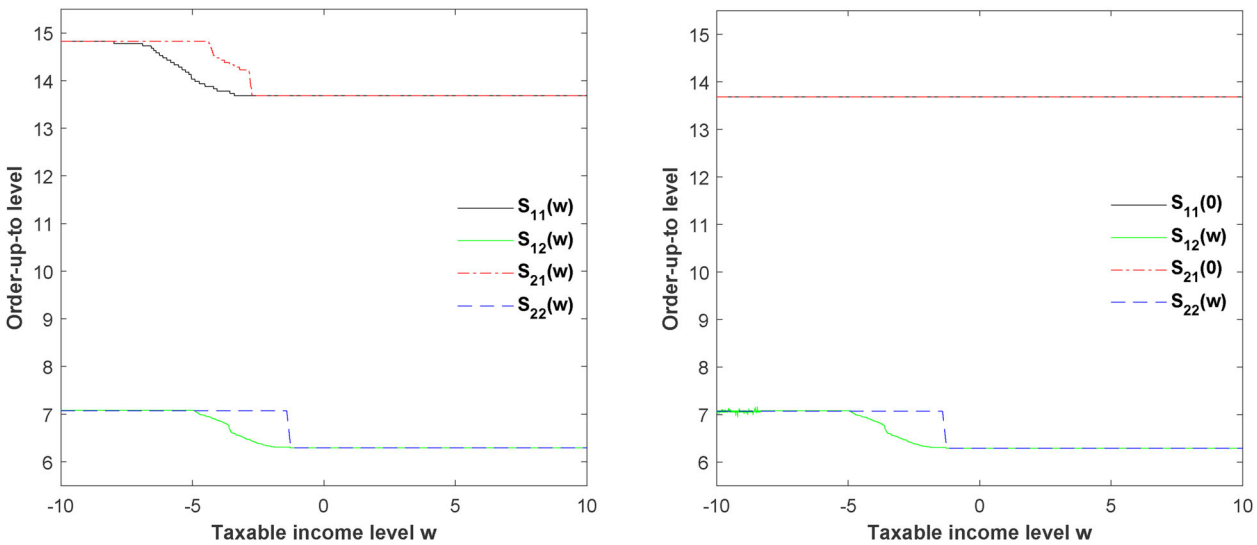
As mentioned earlier, the taxable income for firms with inventory operations should be determined by the accrual accounting method (see, e.g., IRS, 2019). The amount of tax to pay at the end of each accounting period depends on the taxable income the firm generates in the accounting period. Under accrual accounting, the taxable income is equal to the operating income (sales) less the COGS and other operating expenses (e.g., holding cost) where COGS refers to the cost of units that are issued to meet demand. The purchasing cost of the inventory is deducted only when it is sold and the excess inventory is treated as current asset. The taxable income in period (t, m) is

$$\pi_{t,m}(y_{t,m}, D_{t,m}) = (p - c)\min(D_{t,m}, y_{t,m}) - h(y_{t,m} - D_{t,m})^+, \quad 1 \leq t \leq N, 1 \leq m \leq M. \tag{3}$$

To account for loss carryover, we assume that $\eta \times 100\%$ of the losses incurred in each accounting period can be carried forward to the next accounting period where $\eta \in [0, 1]$. For simplicity, we assume that the losses can be carried over during the whole planning horizon and the accumulated losses are automatically counted into the taxable income in each period. The dynamics of the taxable income are therefore expressed as

$$w_{t+1,1} = \gamma^{-1}\eta\min(0, w_{t,M} + \pi_{t,M}(y_{t,M}, D_{t,M}))),$$
$$w_{t,m+1} = w_{t,m} + \pi_{t,m}(y_{t,m}, D_{t,m}), 1 \leq t \leq N, 1 \leq m < M.$$

The firm pays tax at the end of each accounting period. Let $T(w)$ be the amount of tax to be paid when the taxable



(a) Full loss carryover (b) No loss carryover

Figure 1. Optimal basestock levels $S_{t,m}(w)$ when $h = 0$.

income is w. Without loss of generality, we assume that the tax function $T(w)$ is increasing in the taxable income w with a slope being strictly smaller than one, which ensures that the post-tax income, denoted as $U(w) = w - T(w)$, is increasing in the taxable income w. Clearly, when the tax function $T(\cdot)$ is convex, the after-tax income function $U(\cdot)$ is concave. In the base model, we focus our analysis on a proportional (flat-rate) tax system (e.g., current U.S. system with a flat rate of 21%) such that $T(w) = \lambda w^+$. Note that $T(w)$ satisfies the property of *positive homogeneity* in the sense that $T(aw) = aT(w)$ for all $a > 0$. Convexity and positive homogeneity together imply subadditivity, i.e., $T(w_1 + w_2) \leq T(w_1) + T(w_2)$ for any w_1 and w_2.

The firm's objective is to maximize the expected discounted post-tax profit over the planning horizon where tax is paid at the end of each accounting period. Let $\tilde{V}_{t,m}(x, w)$ be the expected discounted after-tax profit from period (t, m) to the end of the planning horizon with the beginning state (x, w). Note that the initial taxable income level of each accounting period is zero if no loss is carried from previous periods. This stochastic program problem can then be formulated into a cyclic dynamic program as follows:

In the tth accounting period $(t = 1, \ldots, N)$, for $m = 1, \ldots, M - 1$,

$$\tilde{V}_{t,m}(x, w) = \max_{y \geq x} \{E[\tilde{\pi}_{t,m}(y, D_{t,m}) + cx + \tilde{V}_{t,m+1}((y - D_{t,m})^+, w + \pi_{t,m}(y, D_{t,m}))]\}, \tag{4}$$

and for $m = M$ we have

$$\tilde{V}_{t,M}(x, w) = \max_{y \geq x} \{E[\tilde{\pi}_{t,M}(y, D_{t,M}) + cx - T(w + \pi_{t,M}(y, D_{t,M})) + \gamma \tilde{V}_{t+1,1}((y - D_{t,M})^+, \gamma^{-1}\eta \min(0, w + \pi_{t,M}(y, D_{t,M})))]\}, \tag{5}$$

where $\tilde{V}_{N+1,1}(x, w) = \nu(x)$.

We assume that $\nu(x)$ is a non-negative and differentiable function of x with $\nu(0) = 0$ and $\nu(x) - cx$ is (weakly) decreasing and concave in x, which implies that the average unit salvage value of ending inventory is no more than the ordering cost. Note that it is common in the inventory literature (see, e.g., Porteus, 2002) to assume that the unit salvage value of excess inventory is the ordering cost (i.e., $\nu(x) = cx$) or zero (i.e., $\nu(x) = 0$).

It is more convenient to work with the transformed value function $V_{t,m}(x, w) = \tilde{V}_{t,m}(x, w) + w - cx$, which can be seen as the net working capital from period (t, m) onwards, including the total discounted after-tax profit, the accumulated taxable income while suppressing the purchasing value of the on-hand inventory. Using the transformed value functions, the optimality equations (4)-(5) can be rewritten as follows. For $t = 1, \ldots, N$,

$$V_{t,m}(x, w) = \max_{y \geq x} E[V_{t,m+1}((y - D_{t,m})^+, w + \pi_{t,m}(y, D_{t,m}))], \quad m = 1, \ldots, M - 1, \tag{6}$$

$$V_{t,M}(x, w) = \max_{y \geq x} E[g_{t,M}((y - D_{t,M})^+, w + \pi_{t,M}(y, D_{t,M}))], \quad t = 1, \ldots, N, \tag{7}$$

where the terminal value function is $V_{N+1,1}(x, w) = \nu(x) + w - cx$ and

$$g_{t,M}(x, u) = u - T(u) - (1 - \gamma)cx - \eta \min(0, u) + \gamma V_{t+1,1}(x, \gamma^{-1}\eta \min(0, u)). \tag{8}$$

The dynamic programming formulation (6)-(7) generalizes the conventional inventory models. It reduces to the standard inventory models or those that focus on maximizing the pre-tax operating profits when $T(w) = 0$. In the absence of taxation, following the analysis of the standard lost-sales models without fixed cost (see, e.g., Porteus

(2002), Pang (2011) and references therein), we know that a basestock policy is optimal when the lead time is zero. In particular, when $\gamma = 1$ and $\nu(x) = cx$, the optimal policy is a myopic basestock policy. In the presence of taxation, it is interesting to ask whether a basestock policy remains optimal and how taxation affects the inventory decisions.

4. Optimal policy

In the inventory system under taxation, the firm needs to take into account the cumulative taxable income in each accounting period in addition to the inventory level when making the replenishment decision. To characterize the structure of the optimal replenishment policy, we need to investigate the structural properties of the value functions in the optimality equations (6)-(7) and identify the conditions when the concavity can be preserved.

With loss carryover, given the terminal wealth level of an accounting period w_1 and the taxable income from the next accounting period w_2, the tax payment at the end of the second accounting period is $T(\gamma^{-1}\eta\min(w_1, 0) + w_2)$. The composition of an increasing convex function and an increasing concave function may not be convex or concave, which introduces the technical challenge in the structural analysis. In a proportional tax system, we find that the tax function convexity can dominate the concavity induced by loss carryforward, which is shown in the following example.

Example 1 (Proportional Tax). *Consider a proportional tax function $T(w) = \lambda w^+$ with loss carryforward at the rate η and taxable incomes (w_1, w_2) received in two consecutive accounting periods. The total tax payment of the two consecutive accounting periods,*

$$T(w_1) + \gamma T(\gamma^{-1}\eta\min(w_1, 0) + w_2)$$
$$= \lambda(\eta w_1 + \gamma w_2)^+ 1_{\{w_1 \le 0\}} + \lambda(w_1 + \gamma w_2^+) 1_{\{w_1 > 0\}},$$

is piecewise linear and convex in w_1 for any given w_2.

The next proposition shows that the concavity of the profit function can be preserved under dynamic programming and characterizes the optimal inventory policy.

Proposition 1 (Optimal Policy). *Consider a proportional tax with loss carryforward at the rate η. For any period (t, m),*

(a) *$V_{t,m}(x, w)$ is jointly concave in (x, w), decreasing in x, and increasing in w with a slope being in $[1 - \lambda, 1]$.*

(b) *The optimal policy is a state-dependent basestock policy with the basestock levels*

$$S_{t,m}(w) = \arg\max_y \left\{E\left[V_{t,m}((y - D_{t,m})^+, w + \pi_{t,m}(y, D_{t,m}))\right]\right\}, \quad m = 1, \ldots, M-1$$

$$S_{t,M}(w) = \arg\max_y \left\{E\left[g_{t,M}((y - D_{t,M})^+, w + \pi_{t,M}(y, D_{t,M}))\right]\right\}.$$

In each period, if the on-hand inventory level is below the basestock level, the firm will order up to the basestock level; Otherwise, no order will be placed.

(c) *If $h = 0$, $V_{t,m}(x, w)$ is submodular in (x, w) and $S_{t,m}(w)$ is decreasing in w.*

Proposition 1 shows that under a proportional tax with loss carryforward, the joint concavity of the value-to-go functions can be preserved under dynamic programming, which ensures that a state-dependent basestock policy is optimal. Note that the optimal state-dependent basestock policy under taxation reduces to the optimal basestock policy that is independent of the income level in the absence of taxation (i.e., $T(w) = 0$ for any w). For convenience, let $\hat{S}_{t,m}$ denote the optimal basestock levels in the tax-free model.

When $h = 0$, the taxable income in each period $\pi_{t,m}(y, D_{t,m})$ is increasing in y. Since the after-tax profit function $U(\cdot)$ is concave, $U(w + \pi_{t,m}(y, D_{t,m}))$ is submodular in (w, y) for any $D_{t,m}$. Part (c) shows that the submodularity can be preserved under dynamic programming, which implies that inventory and taxable income have a substitutable relationship. As a result, the greater the taxable income level, the lower the optimal basestock level. It is interesting to compare our result to that of Chao *et al.* (2008) who analyze the cash-constrained lost-sales inventory model with zero holding cost. They show the optimality of the state-dependent basestock policy where the state is the cash level, but do not reveal any monotonicity of the basestock level with respect to the cash level.

The result in Part(c) is also illustrated in Figure 1 which considers an inventory system with a proportional tax rate of 20%. The planning horizon has two accounting periods, each of which has two ordering periods. The demands are independent and identically distributed (i.i.d.) over time subject to a lognormal distribution with mean 10 and cv of 1. The other parameters are $p = 1.2, c = 1, h = 0, \gamma = 0.8$, and the terminal value function is $\nu(x) = cx$. In both settings of full loss carryforward ($\eta = 100\%$) and no loss carryforward ($\eta = 0$), the optimal basestock level in each period (weakly) decreases in the taxable income.

To gain more intuition on the monotonicity of the optimal basestock level when $h = 0$, we derive a closed-form solution for the last ordering period when $\nu(x) = cx$. According to (8), we have $g_{N,M}(x, u) = U(u) - (1 - \gamma)cx$, which implies that:

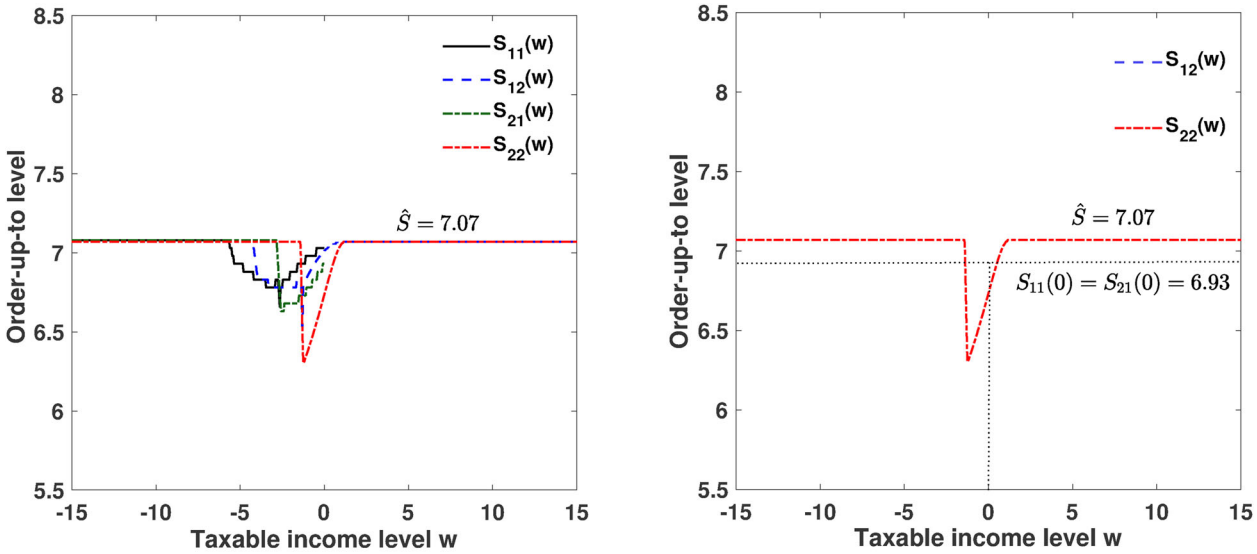
$$E\left[g_{N,M}((y - D_{N,M})^+, w + \pi_{N,M}(y, D_{N,M}))\right]$$
$$= E\left[U(w + (p - c)\min(D_{N,M}, y))\right] - (1 - \gamma)cE\left[(y - D_{N,M})^+\right].$$

Depending on the values of w and y, there are three cases:

- For $w \ge 0$, we have $E[U(w + (p - c)\min(D_{N,M}, y))] = (1 - \lambda)E[w + (p - c)\min(D_{N,M}, y)]$.
- For $w < 0$ and $y \le -\frac{w}{p-c}$, we have $E[U(w + (p - c)\min(D_{N,M}, y))] = E[w + (p - c)\min(D_{N,M}, y)]$.
- For $w < 0$ and $y > -\frac{w}{p-c}$, we have

$$E\left[U(w + (p - c)\min(D_{N,M}, y))\right]$$
$$= E\left[U(w + (p - c)D_{N,M} - (p - c)(D_{N,M} - y)^+)\right]$$
$$= E\left[U((w + (p - c)D_{N,M})^+ - (p - c)(D_{N,M} - y)^+)\right]$$
$$= (1 - \lambda)E\left[(w + (p - c)D_{N,M})^+ - (p - c)(D_{N,M} - y)^+\right],$$

where the last equality is due to the structure of the proportional tax function and that $(w + (p - c)D_{N,M})^+ \ge (p - c)(D_{N,M} - y)^+$ for $y > -\frac{w}{p-c}$.



(a) Full loss carryover (b) No loss carryover

Figure 2. Optimal basestock levels $S_{t,m}(w)$.

Since the objective function is piecewise concave, we can derive the closed-form expression of the basestock level in period (N, M):

$$S_{N,M}(w) = \begin{cases} y_{N,M}(0), & \text{if } w < -(p-c)y_{N,M}(0), \\ -\dfrac{w}{p-c}, & \text{if } -(p-c)y_{N,M}(0) \le w < -(p-c)y_{N,M}(\lambda), \\ y_{N,M}(\lambda), & \text{if } w \ge -(p-c)y_{N,M}(\lambda), \end{cases}$$

where $F_{N,M}(\cdot)$ is the distribution function of $D_{N,M}$ and

$$y_{N,M}(\lambda) = F_{N,M}^{-1}\left(\frac{(1-\lambda)(p-c)}{(1-\lambda)(p-c)+(1-\gamma)c}\right)$$

which is decreasing in λ. Therefore, $S_{N,M}(w)$ is a decreasing function of w.

With positive holding cost (i.e., $h > 0$), it is much more challenging to identify the conditions under which the value functions are submodular in (x, w) and the optimal basestock levels are decreasing in w. The difficulty arises from the fact that $\pi_{t,m}(y, D_{t,m})$ is decreasing in y when $y > D_{t,m}$ for $h > 0$. Therefore, it is unclear whether the function $V_{t,m+1}((y - D_{t,m})^+, w + \pi_{t,m})$ is submodular in (y, w) if $V_{t,m+1}(x, w)$ is jointly concave and submodular in (x, w). The following example shows that the optimal state-dependent basestock level in general is not monotone in the taxable income with positive holding cost.

Example 2 (Optimal basestock levels) *Consider an inventory system with a proportional tax rate of 21%. The planning horizon has two accounting periods, each of which has two ordering periods. Demands are i.i.d. over time subject to a lognormal distribution with mean 10 and cv of one. The other parameters are* $p = 1.2, c = 1, h = 0.2, \gamma = 1$, *and the terminal value function is* $\nu(x) = cx$. *The following figures show the optimal basestock levels in the four ordering periods for both settings with full and no loss carryforward. We observe that the optimal order-up-to level in each period reflects the state-dependent basestock policy structure. With full loss carryforward, Figure 2(a) indicates that the optimal state-dependent basestock levels for all periods have a V-shaped pattern: all of them decrease when the taxable income is sufficiently negative and then increase when the taxable income is closer to zero. With no loss carryforward, Figure 2(b) shows that the optimal state-dependent basestock levels in the second ordering period of both accounting periods are of V-shaped and they coincide. Note that without loss carryforward, the starting taxable income level of each accounting period is always zero and* $S_{11}(0) = S_{21}(0) = 6.93$. *Comparing the optimal basestock level in the second ordering period of the first accounting period,* $S_{12}(w)$, *in both systems, we observe that the optimal basestock levels in both systems coincide when the taxable income level is relatively low or relatively high. The optimal basestock level with full loss carryforward crosses over that with no loss carryforward from below when the taxable income level is negative and close to zero, which implies that the loss carryforward policy may introduce a tax avoidance incentive to drive a lower inventory level when the firm incurs moderate loss. We also compute the optimal basestock level in the absence of taxation, which is a myopic policy. The optimal myopic basestock level* $\hat{S} = 7.07$ *is equal to or greater than the optimal state-dependent basestock levels under taxation.*

Furthermore, it is worth noting that Cohen and Pekelman (1979) consider a setting where each accounting period (i.e., fiscal year) has a single ordering opportunity, which is a special case of our model (i.e., $M = 1$ and without loss carryforward) when the unit ordering cost is constant. Assuming that *the demand is always greater than the beginning inventory level in each period*, they show that a myopic policy is optimal under FIFO. The rationale for such an assumption is the observation that annual inventory turnover (sales/average inventory) is typically greater than one, and hence, it is reasonable to assume that the aggregate annual demand is greater than the initial inventory level of the year. However, such an assumption in general cannot be easily satisfied in stochastic inventory models

and the optimality of a myopic policy requires assumptions on the model parameters (Veinott, 1965; Porteus, 2002). Our analysis does not rely on such an assumption.

We next identify a sufficient condition when a myopic basestock policy is optimal in the setting with a single review period within each accounting period (i.e., $M=1$) and no loss carryforward ($\eta=0$). There is no need to keep track of the taxable income, as it is always zero at the beginning of each accounting period. Therefore, it suffices to use the on-hand inventory to represent the system state. For the ease of exposition, we remove the time index for the ordering period m. The optimality equations become

$$V_t(x) = \max_{y\geq x} E\big[U(\pi_t(y,D_t)) - (1-\gamma)c(y-D_t)^+ + \gamma V_{t+1}((y-D_t)^+)\big], \quad t=1,\ldots,N, \tag{9}$$

with the terminal function $V_{N+1}(x) = \nu(x) - cx$. We next show that a myopic policy is optimal when the system is stationary and the ending inventory is salvaged at the production cost.

Corollary 1 (Myopic policy) *In a system where each accounting period consists of one ordering period (i.e., $N > 1$, $M = 1$) and there is no loss carryforward, a basestock policy is optimal. In addition, if the system is stationary (i.e., demands are i.i.d. and the price and cost parameters are identical across periods) with the terminal function $\nu(x) = cx$, then a myopic basestock policy is optimal with the optimal basestock level*

$$S = \arg\max_{y\geq 0} E\big[U(\pi(y,D)) - (1-\gamma)c(y-D)^+\big]. \tag{10}$$

Corollary 1 provides a sufficient condition when a myopic policy is optimal. Furthermore, in a stationary system where the salvage value of the leftover inventory equals to the ordering cost, we explicitly characterize the myopic basestock level. Under the myopic policy, if the initial inventory level of the planning horizon is below the optimal basestock level, then the starting inventory level is always below the optimal basestock level, and hence, the firm will order up to the optimal basestock level in each period. Otherwise, the initial inventory level of the planning horizon is above the optimal basestock level, the firm will wait until the on-hand inventory level falls below the optimal basestock level in some period and then always order up to the optimal basestock level in the following periods.

Remark 1 (Payment Terms) Many supply contracts allow buyers to make payment within certain length of time after their orders are delivered as a form of trade credit, which is called payment terms. For example, "Net 30", one of the most common payment terms, means that a customer has 30 days to pay the total amount of invoice. Our model can be extended to incorporate payment terms. Assume that the payment term is ℓ periods, i.e., the ordering cost is paid ℓ periods later. We can modify the optimality equations (6)-(7) as:

$$\begin{aligned} V_{t,m}(x,w) &= \max_{y\geq x}\{(1-\gamma_{m+\ell})cy - (1-\gamma_{m+1+\ell})cE\big[(y-D_{t,m})^+\big] \\ &\quad + E\big[V_{t,m+1}((y-D_{t,m})^+, w+\pi_{t,m}(y,D_{t,m}))\big]\}, \quad m<M, \\ V_{t,M}(x,w) &= \max_{y\geq x}\{(1-\gamma_{M+\ell})cy \\ &\quad + E\big[g_{t,M}((y-D_{t,M})^+, w+\pi_{t,M}(y,D_{t,M}))\big]\}, \end{aligned}$$

where the terminal value function is $V_{N+1,1}(x,w) = \nu(x) + w - c_{1+\ell}x$, $\gamma_{m+\ell} = \gamma^{\lfloor\frac{m+\ell-1}{M}\rfloor}$ is the cumulative discount rate of ℓ periods from period (t, m), and

$$\begin{aligned} g_{t,M}(x,u) &= u - T(u) - (1-\gamma_{M+1+\ell})cx - \eta\min(0,u) \\ &\quad + \gamma V_{t+1,1}(x, \gamma^{-1}\eta\min(0,u)). \end{aligned}$$

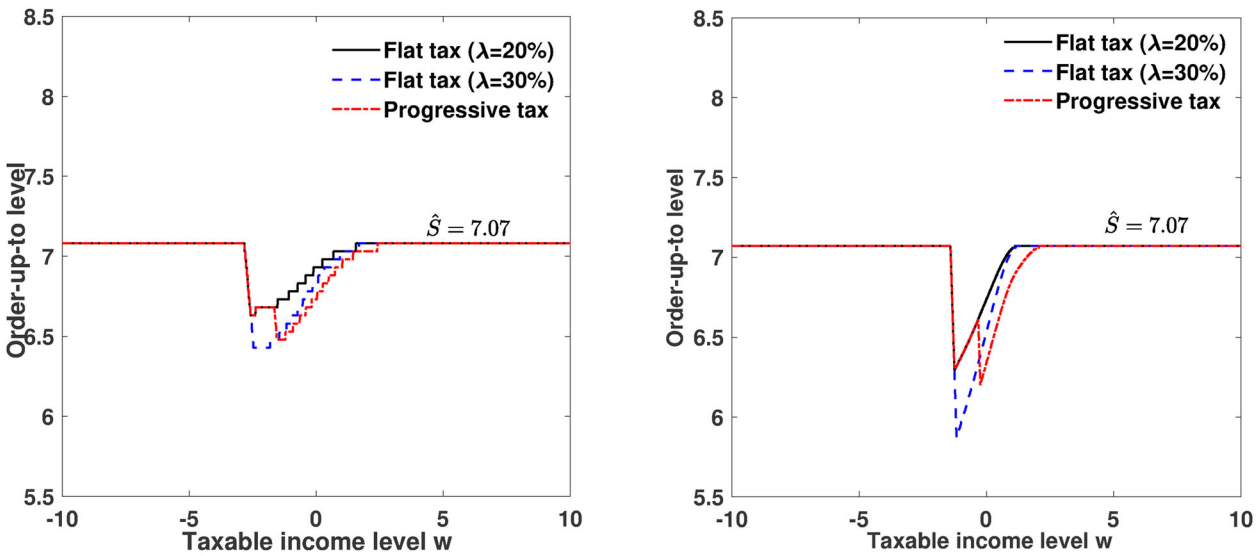
In particular, if $\ell = 0$, $\gamma_{m+\ell} = \gamma^{\lfloor\frac{m-1}{M}\rfloor} = 1$ for all $1 \leq m \leq M$ and $\gamma_{M+1+\ell} = \gamma$, and the model reduces to the one characterized by (6)-(7). Following similar arguments, we can derive the same structural results as Proposition 1.

5. Inter- and intra-accounting-period effects of taxation

What are the effects of taxation on inventory decisions? Example 2 shows that the optimal state-dependent basestock levels under taxation tend to be smaller than the ones in the absence of taxation. In this section, we first provide an economic interpretation on this observation, and then analytically examine the effects of taxation on the inventory decisions.

It is known that tax function convexity may incentivize firms to reduce variations of taxable incomes to reduce expected tax liabilities, which is similar to the risk-averse behavior (Smith and Stulz, 1985; Graham and Smith, 1999). This is because the increasing and concave after-tax profit function $U(\cdot)$ plays a role of a risk-averse utility function. A risk-neutral firm under a convex tax is expected to behave similarly to a risk-averse decision maker. As well established in the decision theory and supported by experimental results, the ordering decisions made by the risk-averters systematically deviate from those made by the risk-neutral decision makers (Agrawal and Seshadri, 2000). However, most of the models focus on single-period settings. For example, Eeckhoudt *et al.* (1995) show that a risk-averse newsvendor with recourse options tends to order less than does a risk-neutral one. It is natural to expect that the optimal basestock levels under a progressive (convex) tax tend to be lower than the ones without taxation. Although the loss carryforward may introduce some risk-taking incentives into the decision making, as we show in Example 1, it is possible that the tax function convexity dominates the effect of loss carryforward. For tractability, we restrict the following analysis to the systems without loss carryforward.

There are two types of intertemporal effects of CIT on dynamic inventory decisions. First, CIT can affect the inventory decisions across different accounting periods that are subject to different tax liabilities, which we term as the *inter-accounting-period effect of taxation*. Second, within each accounting period, the year-end tax affects multiple intra-accounting-period ordering decisions, which we term as the *intra-accounting-period effect of taxation*. Simultaneously



(a) Basestock levels in period (2,1)

(b) Basestock levels in period (2,2)

Figure 3. Effect of taxation for different tax structures.

analyzing the two effects in a dynamic model, if possible, can be very challenging. For tractability, we consider the following two settings: multiple accounting periods where each accounting period has only one operating period to identify the *inter-accounting-period effect of taxation*, and a single accounting period with multiple operating periods to identify the *intra-accounting-period effect of taxation*.

We first study the inter-accounting-period effect of tax on the firm's inventory decisions. To separate from the intra-accounting-period effect, we consider the setting in which each accounting period has one ordering opportunity (i.e., $M = 1$). As we have shown in Corollary 1, a basestock policy is optimal. For convenience, we drop the subscript m for the ordering period, and keep the subscript t for the accounting period. The next proposition compares the firm's optimal basestock levels with and without tax.

Proposition 2 (Inter-Accounting-Period Effect) *Consider a proportional tax system with multiple accounting periods, each of which consists of one review period, and without loss carryover. The optimal basestock levels with and without tax satisfy $S_t \leq \hat{S}_t$. In particular, when the system is stationary and $\nu(x) = cx$, a myopic basestock policy is optimal with $S \leq \hat{S} \equiv F^{-1}\left(\frac{p-c}{p-\gamma c+h}\right)$.*

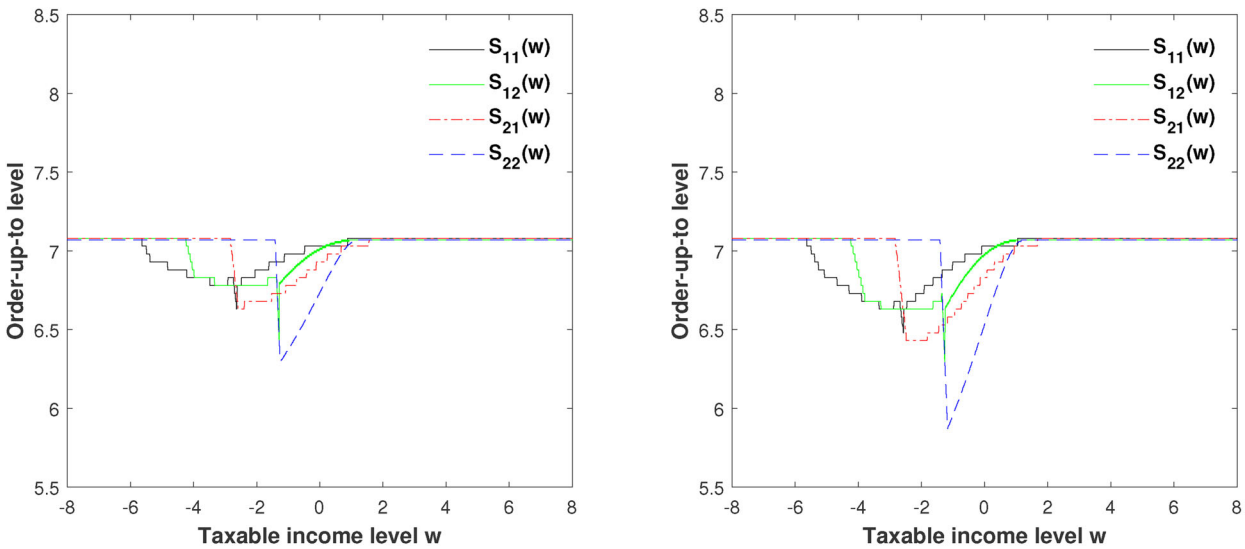
Proposition 2 shows that the firm orders no more than its counterpart without tax considerations in a multi-accounting period model where each accounting period has one ordering opportunity. In this case, given the order-up-to level y, the firm's taxable income in each accounting period satisfies $\pi_t(y, \xi') \leq (p-c)y = \pi_t(y, \xi'')$ for any $\xi' \leq y \leq \xi''$, i.e., the firm's taxable income in the overage case is no greater than that in the underage case. A convex tax implies that the firm pays less tax in the overage case than that in the underage case. Therefore, the potential loss in the after-tax profit due to leftover inventory is less significant than that due to a stockout. In addition, since only the production or procurement cost of the units to satisfy demand will be deducted in the taxable income under accrual accounting, the firm has no incentive to stock excess inventory and carry over to the next accounting period. Consequently, the firm with tax considerations prefers stockouts over leftover inventory and sets an optimal order-up-to level no greater than the one without tax considerations.

We next examine the intra-accounting period effect of tax by considering a planing horizon of a single accounting period (i.e., $N = 1$). For convenience, we drop the subscript for the accounting period t, and keep the subscript of the ordering period m.

Proposition 3 (Intra-Accounting-Period Effect) *Consider a proportional tax system with a single accounting period consisting of multiple ordering periods. If $p \leq c + h$, the optimal basestock levels with and without tax satisfy $S_m(w) \leq \hat{S}_m$ and $S_m(w - (p-c)\delta) \leq S_m(w) + \delta$ for any $\delta > 0$.*

Proposition 3 shows that if the sales margin of a unit is no greater than its holding cost, then the optimal basestock level under a proportional tax is no greater than its counterpart without tax in each period. Furthermore, the marginal value of initial taxable income is lower when part of the inventory is liquidated at the sales margin. In more general settings, although it is technically intractable to compare the optimal order-up-to level with and without tax in each period of an accounting period, a firm under a proportional tax in general orders no more than its no-tax counterpart based on the extensive numerical studies we have conducted.

Propositions 2 and 3 reveal the two types of intertemporal effects of taxation on the dynamic inventory decisions. Although the two effects are identified separately in different special settings due to tractability, since both result in smaller optimal basestock levels, one would expect that their joint effect leads to similar results in more general settings (i.e., multiple accounting periods each of which has multiple ordering periods). We provide a numerical example to further demonstrate the effects of taxation on the optimal inventory decisions in a more general setting.



(a) Basestock levels with flat tax ($\lambda = 20\%$) (b) Basestock levels with flat tax ($\lambda = 30\%$)

Figure 4. Effect of taxation on basestock levels in each period.

Example 3 (Effects of Taxation) *Consider the same setting as Example 2 under three tax systems: (i) a flat tax at the rate $\lambda = 20\%$, (ii) a flat tax at the rate $\lambda = 30\%$, and (iii) a two-bracket progressive tax $T_2(w) = \lambda_1 w^+ + (\lambda_2 - \lambda_1)(w - \tau_2)^+$ with $(\lambda_1, \lambda_2) = (20\%, 30\%)$ and $\tau_2 = 1$. Figure 3 (a) and (b) demonstrate the optimal basestock levels of the two ordering periods in the last accounting period. One can observe that the optimal basestock levels under taxation are no greater than the optimal basestock levels without taxation ($\hat{S} = 7.07$) in both ordering periods. Note that changing from the 20% flat tax system to the 30% flat tax or the two-bracket progressive tax system not only increases the tax rate, but also increases the degree of tax function convexity. Therefore, the optimal basestock levels under a flat tax of 20% are higher than that under the flat tax of 30% and the two-bracket progressive tax. However, when comparing the 30% flat tax to the two-bracket progressive tax system, the latter has a smaller tax function convexity below the threshold τ_2 and a greater convexity above the threshold, which explains that the optimal basestock level under the 30% flat tax crosses over that under the two-bracket progressive tax from above.*

Figure 4(a) and (b) show the joint intra- and inter-accounting-period effects of tax on the basestock levels under the proportional tax in this example. There is only intra-accounting-period effect of tax in the last accounting period (i.e., period (2, 1) and (2, 2)) while there are both inter- and intra-accounting-period effect in the first accounting period (i.e., period (1, 1) and (1, 2)). Comparing the basestock levels in the same review period of different accounting periods, we can see that the joint inter- and intra-accounting-period effects lead to lower basestock levels when the taxable income is relatively small and higher basestock levels when the taxable income is relatively high.

6. The value of tax considerations

Our analysis has shown that inventory management under taxation is much more sophisticated than that without taxation. It is natural to ask whether one can apply the optimal inventory policy derived from the model without considering taxation to approximate the optimal inventory policy in the system with taxation. Our numerical study aims to address the following questions. What is the value of considering taxation in inventory decisions? More specifically, what is the potential loss of the after-tax profit if the firm makes inventory decisions based on the model without considering taxation (i.e., use the expected pre-tax profit as the objective function) despite that it operates under CIT? How do demand uncertainty and the characteristics of the tax system, such as the tax rate and loss carryforward, influence the value of tax considerations in an inventory system?

To address these questions, we consider four settings with different planning horizons: (i) one accounting period consisting of only one ordering period; (ii) two accounting periods each of which consists of only one ordering period; and (iii) one accounting period consisting of two ordering periods; and (iv) two accounting periods each of which consists of two ordering periods. We use $(N, M) = (1, 1)$, (2, 1), (1, 2), and (2, 2) to indicate these settings, respectively.

In each setting, we adopt the same cost and demand parameters as those in the previous numerical examples, i.e., $p = 1.2, c = 1, h = 0.2, \gamma = 1, \nu(x) = cx$. Each ordering period has i.i.d. log-normal demand with mean 10. For both systems with and without loss carryforward, we vary the tax rate λ among the values $\{20\%, 30\%, 40\%, 50\%\}$ under a flat tax $T(w) = \lambda w^+$ and the cv of each ordering period among the values $\{1, 3, 5, 10\}$. Both of the settings with no loss carryforward and full loss carryforward are considered. Furthermore, for ease of comparison, we assume that the starting state in the beginning of the planning horizon is (0, 0).

To measure the profit loss due to ignoring taxation in inventory decisions, we first compute the optimal basestock policy in the model under taxation and its counterpart in the model without considering taxation in each setting. We then calculate the expected pre- and after-tax profits under the optimal basestock policies derived from both models. Let V^{pretax} and $V^{aftertax}$ be the expected discounted pre- and after-tax profits throughout the planning horizon under the optimal policy of the model under taxation, and $\hat{V}^{pretax}$ and

Table 1. The effect of taxation.

Proportional tax systems without loss carryforward ($\eta = 0$)														
(N, M) (%)		*(1, 1)*				*(2, 1)*			*(1, 2)*			*(2, 2)*		
λ	cv	$\hat{S}$	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)
20	1	7.07	6.74	0.63	0.22	6.74	1.26	0.22	6.93	1.29	0.33	6.93	2.59	0.33
	3	3.16	2.75	0.19	1.44	2.75	0.38	1.44	2.93	0.41	1.21	2.93	0.84	1.20
	5	1.96	1.64	0.10	2.32	1.64	0.21	2.32	1.78	0.23	2.06	1.78	0.46	2.04
	10	0.99	0.79	0.04	3.48	0.79	0.09	3.48	0.88	0.10	2.19	0.88	0.20	2.14
30	1	7.07	6.53	0.54	0.61	6.53	1.09	0.61	6.83	1.13	0.46	6.83	2.26	0.46
	3	3.16	2.53	0.16	4.10	2.53	0.32	4.10	2.78	0.35	2.21	2.78	0.72	2.20
	5	1.96	1.47	0.08	6.64	1.47	0.17	6.64	1.68	0.19	3.61	1.68	0.39	3.59
	10	0.99	0.69	0.03	10.29	0.69	0.07	10.29	0.83	0.08	4.42	0.83	0.17	4.36
40	1	7.07	6.30	0.46	1.39	6.30	0.91	1.39	6.73	0.96	0.72	6.73	1.92	0.72
	3	3.16	2.29	0.13	9.53	2.29	0.25	9.53	2.68	0.30	4.03	2.68	0.60	4.02
	5	1.96	1.30	0.06	15.62	1.30	0.13	15.62	1.58	0.16	6.44	1.58	0.33	6.41
	10	0.99	0.59	0.03	24.84	0.59	0.05	24.84	0.73	0.07	8.58	0.73	0.14	8.49
50	1	7.07	6.02	0.37	2.92	6.02	0.74	2.92	6.63	0.79	1.19	6.63	1.59	1.19
	3	3.16	2.04	0.10	20.41	2.04	0.19	20.41	2.53	0.24	7.29	2.48	0.49	7.27
	5	1.96	1.12	0.05	33.97	1.12	0.10	33.97	1.43	0.13	11.53	1.43	0.26	11.48
	10	0.99	0.49	0.02	55.48	0.49	0.04	55.48	0.68	0.05	16.25	0.68	0.11	16.14
Proportional tax systems with full loss carryforward ($\eta = 100\%$)														
(N, M)		*(1, 1)*				*(2, 1)*			*(1, 2)*			*(2, 2)*		
λ	cv	$\hat{S}$	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)	$S_{11}(0)$	$V_{11}(0,0)$	ρ (%)
20	1	7.07	6.74	0.63	0.22	6.96	1.28	0.15	6.93	1.29	0.33	7.03	2.60	0.29
	3	3.16	2.75	0.19	1.44	2.94	0.40	1.03	2.93	0.41	1.21	3.03	0.85	1.04
	5	1.96	1.64	0.10	2.32	1.78	0.21	1.66	1.78	0.23	2.06	1.83	0.47	1.84
	10	0.99	0.79	0.04	3.48	0.86	0.09	2.43	0.88	0.10	2.19	0.88	0.21	1.91
30	1	7.07	6.53	0.54	0.61	6.90	1.11	0.41	6.83	1.13	0.46	7.03	2.27	0.37
	3	3.16	2.53	0.16	4.10	2.82	0.33	2.80	2.78	0.35	2.21	2.93	0.73	1.72
	5	1.96	1.47	0.08	6.64	1.47	0.18	4.47	1.68	0.19	3.61	1.78	0.40	2.96
	10	0.99	0.69	0.03	10.29	0.69	0.07	6.75	0.83	0.08	4.42	0.83	0.18	3.55
40	1	7.07	6.30	0.46	1.39	6.82	0.93	0.93	6.73	0.96	0.72	6.98	1.94	0.52
	3	3.16	2.29	0.13	9.53	2.29	0.27	6.23	2.68	0.30	4.03	2.88	0.62	2.95
	5	1.96	1.30	0.06	15.62	1.56	0.14	10.02	1.58	0.16	6.44	1.68	0.33	4.96
	10	0.99	0.59	0.03	24.84	0.74	0.06	15.44	0.73	0.07	8.58	0.83	0.15	6.54
50	1	7.07	6.02	0.37	2.92	6.72	0.77	1.90	6.63	0.79	1.19	6.98	1.61	0.78
	3	3.16	2.04	0.10	20.41	2.50	0.21	12.77	2.53	0.24	7.29	2.78	0.50	5.13
	5	1.96	1.12	0.05	33.97	1.43	0.11	20.75	1.43	0.13	11.53	1.63	0.27	8.49
	10	0.99	0.49	0.02	55.48	0.64	0.04	32.64	0.68	0.05	16.25	0.78	0.12	11.97

$\hat{V}^{aftertax}$ be the expected discounted pre- and after-tax profits throughout the planning horizon under the optimal policy of the model without considering taxation, respectively. Here, $V^{aftertax} = V_{11}(0,0)$. If the firm does not take tax into account when making inventory decisions, the optimal inventory policy is the basestock policy where the inventory level in each period is denoted as $\hat{S}_{t,m}$. Given that $\gamma = 1$ and $\nu(x) = cx$, a stationary basestock policy is optimal. Substituting the order-up-to level in each period $y_{t,m} = \max(\hat{S}_{t,m}, x_{t,m})$ into the pre-tax profit function and the after-tax profit function give the expected pre-tax profit $\hat{V}^{pretax}$ and after-tax profit $\hat{V}^{aftertax}$ under the optimal policy when the firm does not take tax into account, respectively. For simplicity, we drop the time index and use $\hat{S}$ to indicate the optimal basestock level for the model without taxation.

To measure the effect of taxation on the profitability of the firm, we define the value of considering taxation (or the relative profit loss due to ignoring taxation) for inventory decisions as the percentage loss of the after-tax profit:

$$\rho = \frac{V^{aftertax} - \hat{V}^{aftertax}}{V^{aftertax}} \times 100\%.$$

That is, in a system with taxation, if a firm adopts the optimal inventory policy derived from the model without considering taxation, compared to the optimal policy based on the model with taxation, the relative loss of the after-tax profit is measured by ρ.

The numerical results are reported in Table 1. We have the following observations.

First, in the absence of taxation, myopic optimal basesotck policies are optimal. The optimal myopic basestock levels, denoted by $\hat{S}$, are $7.07, 3.16, 1.96$ and 0.99, for cv $= 1, 3, 5$ and 10, respectively. We find that the optimal basestock levels under taxation are smaller than the corresponding optimal myopic basestock levels, implying that the risk-averting incentive arising from the convexity of tax function drives lower basestock levels.

We further observe that the basestock levels and the loss ratios of the after-tax profit are sensitive to the cv of the demand. With all other parameters being fixed, the optimal basestock level decreases in the cv of the demand whereas the loss ratio of the after-tax profit increases in the cv of demand, which implies that a greater degree of demand uncertainty amplifies the negative impact of taxation on the firm's profit and increases the relative profit loss of ignoring taxation in inventory decisions. In other words, the greater the demand uncertainty, the greater the value of the model with tax consideration. The results can be explained by the risk-averting incentives induced by the tax function

convexity. A risk-averse firm is sensitive and averse to the variation of cash flows. Increasing demand uncertainty leads to greater variation in the pre-tax profit and thus a higher premium (i.e., effective tax payment) and a lower expected after-tax profit. When demand uncertainty is high, the loss due to ignoring tax liabilities becomes significant. These observations imply that when the demand uncertainty is relatively low, the optimal basestock policy derived from the model without tax considerations can serve as a good heuristic for the model with taxation. However, when the demand uncertainty is relatively high, the firm should take taxation into account when making inventory decisions.

In terms of the effect of tax rate, we observe that the optimal basestock level under taxation decreases in the tax rate in all the cases without loss carryforward and most of the cases with full loss carryforward, while the loss ratio of the after-tax profit increases in the tax rate. A higher tax rate implies a greater degree of tax function convexity and therefore a stronger risk-averting incentive, which leads to greater loss due to ignoring tax in the inventory decisions. When both the demand uncertainty (cv) and the tax rate are sufficiently high, the profit loss due to ignoring tax can be significant (e.g., over 50% for $\lambda = 50\%$ and cv $= 5$ in the setting (1, 1)).

Comparing the instances with and without loss carryforward, we find that loss carryforward leads to higher basestock levels and smaller loss ratios of the after-tax profit. Furthermore, the greater the tax rate and the value of cv, the larger the difference between the two systems with and without loss carryover. This can be explained by the fact that loss carryforward allows the firm to spread the losses over time and therefore smooth out the profits over time. As discussed earlier, loss carryforward introduces the risk-taking incentives, which reduces the risk-averting incentives, and therefore, the firm becomes less sensitive to the tax function convexity. Nevertheless, the effect of tax function convexity (risk-averting incentives) cannot be fully offset by the risk-taking incentives induced by loss carryforward. When the tax rate and cv are sufficiently large, the loss due to ignoring taxation can still be sizable.

In summary, our numerical study shows that taxation can have a significant impact on the firm managing an inventory system under demand uncertainty. A high tax rate (high degree of tax function convexity) and high demand uncertainty may lead to significant losses if the firm ignores taxation when making inventory decisions. The loss carryforward policy, though relieves the tax burden, cannot fully revert the effect of tax function convexity.

7. Conclusion and discussions

CIT is one of the most important economic levers to provide incentives for corporate operations and investment decisions and it is often the focus of economic reforms of policy makers such as the 2017 TCJA. In this article, we study a periodic-review inventory system under corporate income tax and capture several key characteristics of the tax systems: accrual accounting method for inventory systems, tax function convexity under a proportional tax, and loss carryforward. We propose a general stochastic dynamic programming formulation that consists of multiple accounting periods, each of which consists of multiple review periods. We show that a state(taxable-income)-dependent basestock policy is optimal for a lost-sales system. We find that the optimal basestock levels are not monotone in the taxable income in general. When the holding cost is zero, a greater taxable income level leads to a lower basestock level, which indicates the substitutable relationship between the inventory level and taxable income. We then study the effects of taxation on the optimal inventory decisions and identify two types of intertemporal effects of taxation, the intra-accounting-period effect and the inter-accounting-period effect of taxation. We show that both effects drive smaller optimal basestock levels in the absence of loss carryforward, which implies that the CIT may induce risk-averting behavior while the loss carryforward may induce risk-seeking behavior. Our numerical study shows that the value of considering tax in inventory decisions can be significant when the tax rate and demand volatility are sufficiently high. The conflicting incentives introduced by taxation reveal the complexity of the effects of taxation on dynamic inventory decisions, which emphasize the importance of investigating dynamic inventory management under taxation. We also extend our analysis to the system under a general progressive tax, the backordering setting, and the cash accounting method.

Our model provides a more general framework for manufacturing firms to better manage their inventories in response to tax reforms and for policy makers to better understand manufacturing activities. Our results can serve as a theoretical foundation to empirically investigate the impact of tax reforms on firms' production and inventory decisions.

Note that our model assumes that the ordering costs are constant over time, which implies that the COGS is independent of the inventory costing method (FIFO or LIFO). If, otherwise, inventory cost changes over time, firms need to track the inventories of different ages to correctly calculate the costs of good sold, which are deducted from the taxable income. The age-dependent nature of the model under FIFO or LIFO leads to a higher-dimensional state space, which is similar to that of perishable inventory models (Nahmias, 2011). Note that Chen *et al.* (2014) find that the structure of the optimal policy of some perishable inventory models can be partially characterized with the notion of $L^\natural$-convexity which reveals some complementary relationship between partial inventory pipelines. However, such structural results may not exist in the presence of tax due to the nonlinearity of tax functions. Cohen and Pekelman (1979) analyze the optimal inventory policies under FIFO and LIFO in a lost-sales system under a flat-rate tax under two critical assumptions as we discussed in detail in literature review. These assumptions, although greatly simplify the analysis, are somewhat restrictive. Dynamic inventory management with taxation under different costing methods remains an interesting and challenging problem.

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